

MTH209: Assignment 8

Below is a solution of Question 2 from exam, taken from a single student's submission. Such a type of solution was given by many students:

```
comp_alpha <- function(alpha, n.vec = c(10, 100, 500, 1000), reps)
{
  l <- length(n.vec)

  #to store mse estimated
  mse1 <- numeric(length = l)
  mse2 <- numeric(length = l)

  # loop for each n in n.vec
  for(i in 1:l)
  {
    m1 <- numeric(length = reps) #to store T1
    m2 <- numeric(length = reps) #to store T2
    n <- n.vec[i]

    # loop for reps
    for(r in 1:reps)
    {
      x <- rgamma(n, alpha, 1)
      m1[r] <- mean(x)
      m2[r] <- (sum((x - m1[r])^2))/(n-1)
    }
    mse1[i] <- mean((m1 - alpha)^2)
    mse2[i] <- mean((m2 - alpha)^2)
  }

  # I have removed the plot and returning the two mse
  return(cbind(mse1, mse2))
}
```

I said very clearly in lab that when running simulation studies, you MUST run the replications as the outside loop, as this helps with parallelizations and also reduces errors. Unfortunately, most of you did not follow

this structure.

TASK: Reorganize the correct solution above, so that the replications are in the outside loop.

```
comp_alpha <- function(alpha, n.vec = c(10, 100, 500, 1000), reps)
{
  ...
  ...
  return(cbind(mse1, mse2))
}
```

INSTRUCTIONS: Upload the file with the name `assignment8.R` on helloIITK. Make sure your code only has the function `comp_alpha` and nothing else.